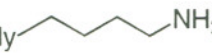
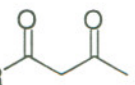
The general use of enzymes for organic synthesis is limited by substrate selectivity in a given reaction. Enzymes generally are highly specific for a particular substrate. Ideally, an enzyme with high catalytic efficiency but low substrate specificity would allow the reaction to be accomplished on a diverse array of substrates increasing its applicability for general use with any desired starting materials.

A recent report by Lerner and coworkers (Barbas III, C.F.; *et. al.*, *Science* **1997**, *278*, 2085-92.) has made tremendous progress toward accomplishing this goal. They have developed a procedure termed "reactive immunization" to grow up and select antibody catalysts in vivo based on their ability to catalyze a particular chemical reaction. This process requires the choice of a hapten (the substrate to which the antibody "binds") which will effectively allow the antibody produced to be optimized for catalyzing the desired chemical reaction.

The aldol reaction was targeted for reactive immunization.

(1) Why would a lysine residue [] be anticipated as the key amino acid in the active site of the antibody produced for catalyzing the aldol reaction? {Justify by showing a key reactive intermediate along the reaction coordinate}

(2) Why was a 1,3-diketone [] an optimal choice for the hapten in their reactive immunization? {Again, justify by showing a key reactive intermediate along the reaction coordinate}

(3) The antibody produced was found to have the catalytic efficiency of natural enzymes, but accepted a much broader spectrum of substrates. The reactive lysine selected in the immunization process was revealed by x-ray to be deeply buried in a hydrophobic pocket at the base of the binding site. Based on the mechanism of catalysis, why would the pKa of the lysine residue have to be significantly perturbed in order for it to efficiently catalyze the aldol at neutral pH?